

STATISTICAL AND MACHINE LEARNING APPROACHES FOR MEDICAL INSURANCE COST PREDICTION

Dissertation Report

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MASTER OF SCIENCE

in

MATHEMATICS

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I, **Abinash Kumar Basa**, Registration Number **324102455**, hereby declare that the work done by me on "**Statistical and Machine Learning Approaches for Medical Insurance Cost Prediction**" under the supervision of **Dr. Mohammad Shuaib, Lovely Professional University, Phagwara, Punjab**, is a record of original work for the partial fulfilment of the requirements for the award of the degree, **Master of Science in Mathematics**. As for the contributions of others are concerned, all efforts have been made to indicate them clearly. The dissertation has also been checked by plagiarism software. If there is any issue regarding originality and plagiarism, the undersigned will be responsible and the awarded degree may be cancelled.

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To Whom-so-ever it may concern

This is to certify that **Abinash Kumar Basa, Registration Number 324102455**, from Lovely Professional University, Phagwara, Punjab, has worked on "**Statistical and Machine Learning Approaches for Medical Insurance Cost Prediction**" under my supervision. It is further certified that the work carried out by the student is a record of original work, to the best of my knowledge, for the partial fulfilment of the requirements for the award of the degree, **Master of Science in Mathematics**. The originality of the research work has also been verified by well-developed plagiarism software.

Name of Supervisor: Dr. Mohammad Shuaib

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List of Abbreviations

ANOVA — Analysis of Variance

BMI — Body Mass Index

GBM — Gradient Boosting Machine

MAE — Mean Absolute Error

ML — Machine Learning

OLS — Ordinary Least Squares

R^2 — Coefficient of Determination

RMSE — Root Mean Square Error

SD — Standard Deviation

SHAP — SHapley Additive exPlanations

SVR — Support Vector Regression

Chapter 1

Introduction

Medical insurance cost prediction has become an important analytical problem at the intersection of statistics, economics, and data science. Health-related expenditure is rarely uniform across a population. Instead, it varies according to demographic, behavioral, and clinical factors, many of which interact in complex ways. For insurers, inaccurate estimation of expected medical charges can weaken pricing strategy and reduce financial efficiency. For policy planners and researchers, poor estimation obscures the underlying relationships between lifestyle variables and cost burden. Because of this, medical cost prediction is not simply a technical forecasting exercise; it is also a way of understanding how risk factors translate into financial consequences.

The present study examines medical insurance charges using both classical statistical inference and machine learning prediction. The problem is well suited to a comparative framework. Statistical methods such as the independent-samples t-test, one-way analysis of variance, Pearson correlation, and multiple linear regression are valuable because they support hypothesis testing and provide interpretable parameter estimates. Machine learning models, by contrast, are often preferred when predictive accuracy is the main objective. A structured comparison between these two traditions therefore provides both explanatory and practical value.

1.1 Background: Healthcare Economics and Insurance Pricing

Health insurance operates on the principle of risk pooling, wherein a large number of individuals contribute premiums so that the collective fund can absorb the relatively unpredictable, and sometimes very large, medical expenses of a smaller subset of policyholders. The financial soundness of this arrangement depends critically on the insurer's ability to estimate, with reasonable accuracy, the expected cost of each policyholder or group of policyholders. This estimation exercise, commonly referred to as actuarial pricing or risk rating, has historically relied on classical statistical techniques applied to demographic and clinical variables such as age, sex, body mass index, smoking status, and family size.

From an economic perspective, mispricing of insurance premiums carries consequences in both directions. If premiums are set too high relative to true expected cost, healthier individuals may

exit the risk pool, a phenomenon known in the insurance literature as adverse selection, leaving the insurer with a progressively costlier and less profitable book of business. Conversely, if premiums are set too low, the insurer faces the risk of under-capitalization and potential insolvency. Both outcomes highlight why the statistical and predictive modelling of medical expenditure is not merely an academic exercise but a matter of direct commercial and regulatory importance.

Beyond the insurer's perspective, healthcare cost modelling is also relevant to public policy. Governments and regulators use aggregate cost models to design subsidy schemes, assess the fiscal burden of public health insurance programmes, and evaluate the likely effect of preventive health interventions, such as anti-smoking campaigns, on long-run healthcare expenditure. A clear quantitative understanding of which behavioral and demographic factors drive costs therefore has value that extends well beyond the insurance industry itself.

1.2 Statistical and Machine Learning Motivation

The dataset used in the present study is the Medical Cost Personal Dataset, containing 1,338 individual records and 7 variables: age, sex, bmi, children, smoker, region, and charges. The dataset contains no missing values, making it suitable for direct modelling after standard preprocessing. The response variable is charges, representing individual medical insurance expenditure. The predictor set includes demographic variables, a lifestyle indicator, a family-size variable, and a regional category. This structure makes it possible to study both inferential relationships and predictive performance in a unified framework.

From a mathematical and applied perspective, insurance cost data present several analytical challenges. The distribution of charges is typically skewed rather than symmetric, meaning that extreme observations can influence summary measures and model fit. Moreover, some predictors may have weak individual effects but become more important when combined with others. Smoking status, for instance, may create a strong shift in expected charges, whereas age and body mass index may produce more gradual increases. The problem therefore requires tools that can identify significance, quantify effect size, and compare predictive efficiency across models with different assumptions.

The motivation for using classical statistical inference lies in its capacity to answer well-defined hypothesis-driven questions: does smoking status significantly affect charges; does region significantly affect charges; and how much of the variance in charges can be explained by a linear combination of the available predictors. These questions are naturally addressed through t-tests, ANOVA, correlation analysis, and ordinary least squares regression, each of which yields not only a point estimate but also a formal significance level that can be reported and defended.

The motivation for using machine learning models is different in character. Rather than testing pre-specified hypotheses about individual variables, machine learning algorithms such as Decision Trees, Random Forests, Gradient Boosting, and Support Vector Regression are optimized purely to minimize prediction error on unseen data. These models can automatically capture non-linear relationships and interaction effects among predictors without the analyst having to specify them in advance. This flexibility often comes at the cost of interpretability, which is why the present dissertation treats interpretability and predictive accuracy as two related but distinct objectives to be evaluated side by side rather than in isolation.

1.3 Research Significance and Mathematical Relevance

The academic relevance of the dissertation lies in its integration of mathematical reasoning with applied machine learning. Rather than treating statistical testing and machine learning as separate domains, the study places them within a common analytical sequence. Significance testing identifies whether group differences and linear relationships exist, while regression and machine learning assess how well charges can be predicted from the same inputs. This combined approach strengthens the analytical coherence of the dissertation.

As a work situated within an M.Sc. Mathematics program, the dissertation places particular emphasis on the underlying mathematical structure of the methods employed. The ordinary least squares estimator is derived from the minimization of a quadratic loss function using matrix calculus; the evaluation metrics used to compare models, namely the coefficient of determination, the root mean square error, and the mean absolute error, are themselves defined through precise algebraic expressions; and the ensemble methods used for prediction, particularly Gradient Boosting, rest on the iterative minimization of a loss functional through functional gradient descent. Presenting these mathematical foundations alongside the applied results allows the

dissertation to remain grounded in the discipline of mathematics while still addressing a concrete, real-world problem.

1.4 Statement of the Problem

The central aim of the present dissertation is to determine which demographic and health-related factors significantly affect medical insurance charges and to compare the predictive performance of selected regression and machine learning models. The study also seeks to evaluate the trade-off between interpretability and accuracy, since the most accurate model is not always the easiest to explain in academic, clinical, or insurance settings. This issue is especially relevant when decision-makers require transparent justification rather than black-box prediction alone.

1.5 Objectives of the Study

The specific objectives of the study are as follows:

1. To statistically test which demographic and health-related variables significantly affect medical insurance charges.
2. To build and compare multiple regression and machine learning models for insurance cost prediction.
3. To evaluate model performance using R-squared, RMSE, and MAE.
4. To identify the most practically useful modelling approach by considering both predictive accuracy and interpretability.

1.6 Hypotheses of the Study

In line with these objectives, the present study tests three hypotheses:

H1: Smoking status has a statistically significant effect on insurance charges. H2: Region has a statistically significant effect on insurance charges. H3: Ensemble and tree-based machine learning models outperform linear regression in predictive accuracy.

1.7 Significance of the Study for Stakeholders

The present study carries distinct significance for three broad categories of stakeholders. For insurers and actuaries, the study offers a concrete demonstration of how a compact set of demographic and behavioral variables can be tested for statistical significance and then used within both an interpretable linear model and a higher-accuracy ensemble model, giving practitioners a

template for deciding how much predictive accuracy they are willing to trade for explanatory transparency in a given regulatory or business context.

For healthcare policy planners, the finding that smoking status alone accounts for the large majority of the explained variance in predicted charges, as shown later in the Gradient Boosting feature-importance results in Chapter 4, provides a quantitative argument in support of preventive health interventions such as anti-smoking program, since even a modest reduction in smoking prevalence within an insured population could be expected to produce a disproportionately large reduction in aggregate claims expenditure.

For students and researchers in applied mathematics and statistics, the dissertation illustrates, within a single coherent dataset, how classical inferential techniques and modern ensemble learning methods can be used side by side rather than as competing paradigms. This is intended to be of pedagogical value beyond the specific findings reported, since the underlying comparative methodology, testing significance first and then benchmarking predictive models against each other using standardized metrics, can be transferred to other cost-prediction or risk-estimation problems in economics, actuarial science, and public health.

1.8 Scope and Limitations of the Study

It is also useful, at the outset, to distinguish the scope of the present study from adjacent research questions that a reader might expect a dissertation on medical insurance cost to address but which fall outside the boundaries defined here. The study does not attempt to build a real-time premium-quoting system, nor does it attempt to model insurer-level profitability or solvency, both of which would require data on claims reserving, reinsurance arrangements, and administrative expense ratios that are not present in the dataset used. Similarly, the study does not attempt fraud detection or anomaly detection in claims, which is a related but methodologically distinct problem typically addressed through classification rather than regression techniques. By keeping the scope narrowly focused on the statistical significance and predictive modelling of individual-level charges from a fixed set of seven variables, the dissertation is able to treat each stage of the analysis, from hypothesis testing through to ensemble model comparison, with the depth appropriate to a mathematics dissertation, rather than attempting a broader but shallower treatment of the wider insurance-analytics domain.

The scope of the study is limited to the variables available in the dataset and to the results obtained from the analyses conducted. The work does not attempt to model all possible determinants of medical spending, such as pre-existing conditions, employment type, medication history, or longitudinal claims behavior. Even with these limits, the study remains useful because it demonstrates how a compact, structured dataset can reveal meaningful cost patterns and support predictive modelling.

1.9 Organization of the Dissertation

The dissertation is organized into five chapters. Chapter 1 introduces the topic, objectives, hypotheses, and scope of the study. Chapter 2 presents a review of relevant literature on healthcare expenditure modelling and insurance cost prediction. Chapter 3 explains the work plan, including the research design, sample, tools, and analytical methods. Chapter 4 reports and discusses the statistical and machine learning results. Chapter 5 summarizes the findings, presents the conclusion, identifies limitations, and suggests future directions.

Chapter 2

Review of Literature

Research on healthcare expenditure has evolved from conventional regression-based explanation toward more flexible predictive modelling. Earlier studies were often concerned with identifying how specific behavioral and demographic variables influence spending, whereas more recent studies have focused on the comparative performance of machine learning algorithms, and, increasingly, on making such models explainable to non-technical stakeholders. The present chapter reviews selected verifiable studies in chronological order in order to situate the present dissertation within that progression.

2.1 Chronological Review

2.1.1 Sturm, An, Maroba, and Patel (2013)

A useful starting point is the work of Sturm, An, Maroba, and Patel (2013), who examined the relationship between obesity, smoking, alcohol intake, and healthcare expenditure in South Africa. Their study analyzed a large cross-sectional sample of 69,380 individuals using survey and medical claims data combined with ordinary least squares regression. The study found that moderate obesity was associated with an 11% increase in medical expenditure, severe obesity with a 23% increase, and current or past smoking with a 13% increase in expenditure. The importance of this study for the present dissertation lies in its clear demonstration that lifestyle variables such as smoking and body mass are economically consequential, not merely clinically descriptive. It supports the view that behavioral risk factors should be treated as major predictors in cost modelling rather than as secondary controls.

2.1.2 Kan, Kharrazi, Chang, Bodycombe, Lemke, and Weiner (2019)

A later development appears in Kan, Kharrazi, Chang, Bodycombe, Lemke, and Weiner (2019), who investigated whether penalized linear regression could improve risk adjustment in predicting healthcare costs for older adults. Using data from 81,106 Medicare Advantage patients observed over multiple years, the authors compared ordinary least squares with lasso, ridge, and elastic net regression. Their results showed that penalized regression performed better than standard OLS, and that lasso regression provided superior prediction ratios across different levels of risk. This

study is important because it shows that linear-model families remain relevant in healthcare cost modelling when enhanced by regularization. It also illustrates that interpretability and predictive performance need not always be treated as opposing goals.

2.1.3 Hassan, Iqbal, Hussain, AlSalman, Mosleh, and Ullah (2021)

The transition to broader machine learning comparison is visible in Hassan, Iqbal, Hussain, AlSalman, Mosleh, and Ullah (2021). Their study, published in *Mathematical Problems in Engineering*, used a medical insurance cost dataset to compare several regression-oriented machine learning methods, including linear regression, support vector regression, ridge regression, stochastic gradient boosting, XGBoost, decision tree, random forest, multiple linear regression, and k-nearest neighbors. The authors reported that stochastic gradient boosting outperformed the competing methods and achieved roughly 86% accuracy, with the study emphasizing the value of computational intelligence for insurance cost prediction. This work is especially relevant to the present dissertation because it frames boosting methods as strong candidates for cost prediction tasks involving non-linear structure and interaction effects.

2.1.4 Cenita, Asuncion, and Victoriano (2023)

Cenita, Asuncion, and Victoriano (2023), in a study anchored on the Knowledge Discovery in Databases process, compared three machine learning regression models, namely Linear Regression, Gradient Boosting, and Support Vector Machine regression, for predicting medical insurance cost. Performance was evaluated using RMSE, R-squared, and k-fold cross-validation. The authors reported that Gradient Boosting achieved the strongest performance among the three models, with an R-squared of approximately 0.892 and an RMSE of approximately 1336.6, and that the k-fold cross-validation results were broadly consistent with the single-split findings. This study strengthens the case, already suggested by Hassan et al. (2021), that gradient boosting methods are consistently competitive on this type of structured insurance-cost data.

2.1.5 Orji and Ukwandu (2024)

Orji and Ukwandu (2023, published in *Machine Learning with Applications* in 2024) deployed three regression-based ensemble machine learning models, namely Extreme Gradient Boosting, Gradient Boosting Machine, and Random Forest, to predict medical insurance costs, and

additionally applied Explainable Artificial Intelligence methods, specifically SHapley Additive exPlanations (SHAP) and Individual Conditional Expectation plots, to identify the key determinants of insurance premium prices. The authors reported that the XGBoost model achieved the strongest overall predictive performance but required greater computational resources, whereas the Random Forest model recorded a somewhat higher prediction error but consumed markedly fewer computing resources. This study is particularly relevant to the present dissertation because it explicitly foregrounds the trade-off between predictive accuracy, interpretability, and computational cost, and because its use of SHAP values illustrates a concrete pathway by which the present study's own ensemble results could be made more transparent in future work.

2.1.6 Albalawi, Alshahrani, Albalawi, Alharbi, and Alhakamy (2023)

Another useful contribution is provided by Albalawi, Alshahrani, Albalawi, Alharbi, and Alhakamy (2023) in Computers and Informatics. Their paper addressed healthcare insurance cost prediction using regression-based machine learning approaches and compared linear regression with polynomial regression on a dataset defined by age, sex, BMI, region, number of children, smoking status, and charges. The study concluded that polynomial regression outperformed linear regression, arguing that the task contains non-linear structure that a purely linear model struggles to capture. The authors also reported strong performance for a gradient-boosted tree regression model in a Spark-based implementation, with R-squared equal to 0.9067. The importance of this work lies in its clear recognition that model flexibility matters when relationships between predictors and expenditure are not strictly linear.

2.2 Comparative Summary of Reviewed Studies

Table 2.1: Chronological Summary of Reviewed Literature

Study	Sample / Data	Methods Used	Key Finding
Sturm et al. (2013)	69,380 (South Africa survey/claims)	OLS regression	Obesity and smoking raise expenditure by 11-23% and 13% respectively
Kan et al. (2019)	81,106 Medicare Advantage patients	OLS, Lasso, Ridge, Elastic Net	Penalized regression outperforms standard OLS
Hassan et al. (2021)	Medical insurance cost dataset	9 ML/regression methods	Stochastic gradient boosting best (~86% accuracy)

Cenita et al. (2023)	Medical insurance cost dataset	Linear Reg., Gradient Boosting, SVM	Gradient Boosting best (R2~0.892)
Orji & Ukwandu (2023/24)	986 records (Kaggle)	XGBoost, GBM, RF + SHAP/ICE	XGBoost best accuracy; RF most efficient
Albalawi et al. (2023)	Medical insurance cost dataset	Linear vs Polynomial vs GB (Spark)	Polynomial > Linear; GB achieves R2=0.9067

2.3 Emerging Themes: Ensemble Methods and Explainable AI

Across the studies reviewed above, several trends become clear. First, smoking and obesity-related variables repeatedly emerge as important cost drivers. Second, linear models remain valuable for explanation and hypothesis testing, particularly when interpretability is needed, and penalized variants of linear regression can narrow the performance gap with more flexible methods. Third, more flexible algorithms, especially gradient boosting variants such as stochastic gradient boosting, Gradient Boosting Machine, and Extreme Gradient Boosting, frequently outperform simpler regression when the objective is prediction, a pattern that recurs across Hassan et al. (2021), Cenita et al. (2023), Orji and Ukwandu (2023/2024), and Albalawi et al. (2023). Fourth, recent literature increasingly evaluates models using explicit accuracy metrics such as R-squared, RMSE, and MAE rather than relying only on coefficient significance. Fifth, and most recently, the literature has begun to incorporate explainability tools, most notably SHAP values, alongside predictive modelling, reflecting a growing recognition that black-box accuracy alone is not sufficient for stakeholders in insurance, healthcare, or regulatory settings.

These developments align closely with the analytical structure of the present dissertation. The recurring dominance of gradient boosting methods across independent datasets and research groups provides converging evidence, external to the present study, that such methods are well suited to the structure of medical insurance cost data; this expectation is subsequently tested directly against the present dataset in Chapter 4.

2.4 Critical Analysis

A critical reading of the six studies reviewed above suggests that the field has progressed through three broad phases. The first phase, represented by Sturm et al. (2013), was primarily concerned with establishing, through classical regression on large survey samples, that specific behavioral

and clinical factors carry an economically significant cost. The second phase, represented by Kan et al. (2019), refined the linear-model tradition itself, showing that regularization techniques such as lasso and ridge regression could materially improve the risk-adjustment performance of what remained, fundamentally, a linear framework. The third and most recent phase, represented by Hassan et al. (2021), Cenita et al. (2023), Orji and Ukwandu (2023/2024), and Albalawi et al. (2023), shifted the emphasis toward comparing an increasingly wide array of machine learning algorithms against each other and against linear baselines, with gradient boosting variants consistently emerging as the strongest performers.

It is notable that despite differences in sample size, ranging from 986 records in Orji and Ukwandu (2023/2024) to 81,106 patients in Kan et al. (2019), and despite differences in geography, timeframe, and modelling software, the qualitative conclusion that boosting-based ensembles outperform plain linear regression on medical cost data recurs with considerable consistency. This convergence across independent studies strengthens, rather than merely illustrates, the third hypothesis tested in the present dissertation. At the same time, a critical limitation shared by nearly all of the reviewed studies, including the present one, is a reliance on a single held-out test split or a small number of cross-validation folds rather than repeated resampling with confidence intervals around the reported R-squared, RMSE, and MAE values; this makes it difficult to judge, from the literature alone, whether the differences between the top two or three models in any given study, including Gradient Boosting versus Random Forest in the present dissertation, are themselves statistically distinguishable or fall within the bounds of ordinary sampling variability. This observation is carried forward as a specific direction for future work in Chapter 5.

2.5 Research Gap

Even so, a gap remains. Much of the published literature either emphasizes prediction without sufficiently discussing interpretability, or focuses on regression explanation without a strong comparative analysis of modern machine learning models. Where explainability is addressed, as in Orji and Ukwandu (2023/2024), it is typically applied only to the best-performing ensemble model rather than used as a basis for comparing the explanatory value of classical statistical inference against the predictive value of machine learning within a single coherent study. The present dissertation addresses that gap by placing statistical inference and predictive modelling in the same study, using one dataset and one coherent evaluation structure. It does not merely ask

which variables are significant, nor does it only ask which model is most accurate. Instead, it asks how the explanatory value of classical methods, namely t-tests, ANOVA, correlation, and OLS regression, compares with the predictive advantage of ensemble and tree-based models, namely Decision Tree, Random Forest, Gradient Boosting, and Support Vector Regression.

The present study is therefore positioned within a literature that increasingly values predictive strength but still requires transparent interpretation. By combining hypothesis testing, correlation analysis, ordinary least squares regression, and machine learning comparison, the dissertation contributes a balanced framework for understanding medical insurance charges.

Chapter 3

Work Plan

3.1 Research Problem

Medical insurance expenditure varies substantially across individuals, yet the relative influence of demographic, behavioral, and regional factors is not uniform. Traditional statistical methods can identify whether variables are associated with expenditure, but they may not always produce the most accurate predictions. Machine learning models often improve predictive performance, though they may reduce interpretability. The research problem addressed in the present study is therefore twofold: first, to determine which variables significantly influence insurance charges; and second, to identify which modelling approach offers the best balance between explanation and prediction.

3.2 Objectives of the Study

The study was guided by the following objectives:

1. To test the significance of demographic and health-related variables in relation to insurance charges.
2. To compare selected statistical and machine learning models for cost prediction.
3. To evaluate the predictive performance of these models using standardized error and fit measures.
4. To recommend the most practically applicable modelling strategy.

3.3 Hypotheses

The study tested the following hypotheses:

H1: Smoking status has a statistically significant effect on insurance charges. H2: Region has a statistically significant effect on insurance charges. H3: Ensemble and tree-based machine learning models outperform linear regression in predictive accuracy.

3.4 Operational Definitions

For analytical clarity, the main variables are defined as follows:

Age: The age of the insured individual in years. Sex: Biological sex category recorded in the dataset. BMI: Body mass index, a standard weight-for-height indicator, computed as weight in kilograms divided by the square of height in meters. Children: Number of dependent children covered by the insurance plan. Smoker: Smoking status of the insured individual. Region: Residential region category in the dataset. Charges: Individual medical insurance cost, used as the dependent variable.

3.5 Sample and Dataset

The study used the Medical Cost Personal Dataset, consisting of 1,338 records and 7 variables: age, sex, bmi, children, smoker, region, and charges. The dataset contains no missing values. The data source is the dataset accompanying Brett Lantz's Machine Learning with R.

The sample may be described as secondary quantitative data suitable for correlational and predictive analysis. Because the dataset contains structured tabular variables, it supports both inferential testing and supervised regression modelling.

3.6 Research Design

The present study adopted a quantitative, correlational, and predictive research design. The inferential component examined whether variables such as smoking status and region are significantly related to charges. The predictive component assessed whether different modelling approaches can accurately estimate charges using the available explanatory variables.

This design is appropriate because the study does not manipulate variables experimentally. Instead, it analyses observed data and evaluates patterns already present in the sample. The focus is therefore analytical rather than experimental.

3.7 Tools and Software

The study used Python with the following libraries:

pandas, NumPy, scikit-learn, stats models, and SciPy.

These tools are suitable because they collectively support data manipulation, hypothesis testing, regression estimation, and machine learning evaluation.

3.8 Data Analysis Techniques

The study employed the following techniques:

1. Descriptive statistics to summarize the distribution of insurance charges.
2. Independent-samples t-tests to compare mean charges across two-category variables such as smoker versus non-smoker, and male versus female.
3. One-way ANOVA to test differences across multi-category variables such as region and number of children.
4. Pearson correlation to quantify the linear relationship between predictor variables and charges.
5. Multiple linear regression (OLS) to estimate the joint effect of predictors on charges and assess explanatory power.
6. Machine learning regression models including Decision Tree, Random Forest, Gradient Boosting, and Support Vector Regression.
7. Model evaluation metrics, namely R-squared, RMSE, and MAE.
8. An 80-20 train-test split, with random state = 42, for predictive model comparison.

3.9 Mathematical Formulation of Statistical Methods

3.9.1 Independent-Samples t-Test

For two independent groups with sample means $\bar{x}_1$ and $\bar{x}_2$, sample variances s_1^2 and s_2^2 , and sample sizes n_1 and n_2 , the test statistic under the assumption of unequal variances (Welch's t-test) is given by:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Also the Welch–Satterthwaite degrees of freedom equation is given by:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

The resulting t statistic is compared against the Student's t-distribution with degrees of freedom computed via the Welch-Satterthwaite equation, and the associated p-value indicates whether the observed difference in group means is unlikely to have arisen by chance alone.

3.9.2 One-Way Analysis of Variance

For a categorical variable with k groups, one-way ANOVA partitions the total sum of squares (SST) into a between-groups sum of squares (SSB) and a within-groups sum of squares (SSW):

$$SST = SSB + SSW$$

$$SSB = \sum_{i=1}^k n_i (\bar{x}_i - \bar{x})^2, \quad SSW = \sum_{i=1}^k \sum_{j=1}^{n_i} (x_{ij} - \bar{x}_i)^2$$

The F statistic is then defined as:

$$F = (SSB / (k - 1)) / (SSW / (N - k))$$

where N is the total number of observations. A large F value, associated with a small p-value, indicates that the group means differ by more than would be expected under the null hypothesis of equal population means.

3.9.3 Pearson Correlation Coefficient

The Pearson correlation coefficient between a predictor variable X and the response variable Y (charges) is defined as:

$$r = \frac{\text{Cov}(X, Y)}{s_X s_Y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

The coefficient r lies between -1 and +1, with values close to the extremes indicating a strong linear association and values close to zero indicating a weak linear association.

3.9.4 Ordinary Least Squares Regression

Multiple linear regression models the response variable as a linear combination of predictors plus an error term:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\mathbf{y}$ is the $n \times 1$ vector of charges, $\mathbf{X}$ is the $n \times p$ design matrix of predictors (including a constant column for the intercept), $\boldsymbol{\beta}$ is the $p \times 1$ vector of unknown coefficients, and $\boldsymbol{\epsilon}$ is the $n \times 1$ vector of residuals, assumed to have zero mean and constant variance.

The ordinary least squares estimator minimizes the residual sum of squares:

$$RSS(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Differentiating RSS with respect to $\boldsymbol{\beta}$ and setting the derivative to zero yields the normal equations:

$$\mathbf{X}^T \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^T \mathbf{y}$$

so that the closed-form OLS estimator is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

The overall goodness of fit is measured by the coefficient of determination:

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

and the adjusted R-squared, which penalizes the inclusion of additional predictors that do not improve the fit, is given by:

Adjusted $R^2 =$

$$R_{\text{adj}}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

The overall significance of the regression model is tested using the F-statistic:

$$F = \frac{SS_{\text{reg}}/p}{SS_{\text{res}}/(n - p - 1)}$$

while the significance of each individual coefficient β_j is tested using its own t-statistic, computed as the coefficient divided by its estimated standard error.

3.9.5 Evaluation Metrics for Predictive Models

Three metrics were used to evaluate and compare the predictive performance of the regression and machine learning models on the held-out test set:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

RMSE penalizes large errors more heavily than MAE because errors are squared before averaging, which makes RMSE particularly sensitive to the small number of very high-cost individuals present in a right-skewed distribution such as insurance charges. MAE, by contrast, treats all errors proportionally and is therefore more robust to a small number of extreme residuals. Reporting both metrics alongside R-squared gives a more complete picture of model performance than any single metric in isolation.

3.10 Mathematical Basis of the Machine Learning Models

3.10.1 Decision Tree Regression

A regression tree partitions the predictor space into a set of disjoint regions $R_1, R_2, \dots, R_M$ and predicts a constant value, typically the mean of the training observations, within each region. At each split, the algorithm selects the predictor and split point that minimize the sum of squared residuals across the two resulting child nodes:

minimize over (j, s):

$$\min_{j,s} \left[\sum_{x_i \in R_1(j,s)} (y_i - \bar{y}_{R_1})^2 + \sum_{x_i \in R_2(j,s)} (y_i - \bar{y}_{R_2})^2 \right]$$

where j indexes the candidate splitting variable and s indexes the candidate split threshold. This process is applied recursively until a stopping criterion, such as a minimum number of observations per leaf or a maximum tree depth, is reached.

3.10.2 Random Forest Regression

Random Forest builds an ensemble of B decision trees, each trained on a bootstrap resample of the training data, with each split additionally restricted to a random subset of the available predictors. The final prediction is the average of the individual tree predictions:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B T_b(x)$$

Averaging over many de-correlated trees reduces the variance of the overall prediction relative to a single decision tree, while the random restriction of predictors at each split reduces the correlation between the individual trees, further improving the stability of the ensemble.

3.10.3 Gradient Boosting Regression

Gradient Boosting builds an additive ensemble of weak learners, typically shallow regression trees, in a stage-wise manner. Given a differentiable loss function $L(y, F(x))$, such as the squared-error loss, the model is updated at each iteration m by fitting a new tree $h_m(x)$ to the negative gradient (pseudo-residuals) of the loss function with respect to the current model predictions:

$$r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F(x)=F_{m-1}(x)}$$

The model is then updated as:

$$F_m(x) = F_{m-1}(x) + \nu h_m(x)$$

where ν is a learning rate, typically a small positive value less than 1, that controls the contribution of each successive tree and helps to prevent overfitting. Because each new tree is fitted to correct the errors of the ensemble built so far, Gradient Boosting can capture complex, non-linear interactions among predictors that a single tree or a linear model would miss.

3.10.4 Support Vector Regression

Support Vector Regression seeks a function $f(x) = w^T x + b$ that deviates from the observed targets y_i by at most a margin ϵ for as many training points as possible, while keeping the function as flat as possible. This is formalized as the optimization problem:

$$\min_{w, b, \xi, \xi^*} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*)$$

Subject to:

$$y_i - (w^T x_i + b) \leq \epsilon + \xi_i$$

$$(w^T x_i + b) - y_i \leq \epsilon + \xi_i^*$$

$$\xi_i, \xi_i^* \geq 0$$

with slack variables introduced to allow limited violations of the margin. Support Vector Regression can be extended to non-linear relationships through the use of kernel functions, such as the radial basis function kernel, which implicitly map the input features into a higher-dimensional space in which a linear relationship may hold. Support Vector Regression is sensitive to the scaling of input features and to the choice of kernel and hyperparameters, which is a consideration revisited when interpreting its comparatively poor performance in Chapter 4.

3.10.5 Worked Illustration of the OLS Normal Equations

To make the derivation in Section 3.9.4 concrete, consider a simplified illustrative regression of charges on two standardized predictors, smoker status (coded 1 for smoker, 0 otherwise) and age, using a small representative subset of observations. The design matrix X would take the form of a column of ones (for the intercept), a column of smoker indicators, and a column of ages. Under the normal equations, $X^T X$ is a 3 x 3 matrix whose diagonal entries are the sums of squares of each column and whose off-diagonal entries are the corresponding cross-products, while $X^T y$ is a 3 x 1 vector containing the sum of charges, the sum of charges among smokers, and the sum of the products of age and charges. Solving $X^T X \beta = X^T y$ for β by matrix inversion, or equivalently by Gaussian elimination, yields the estimated intercept and slope coefficients reported in Table

4.9. This illustrates, in miniature, the same computation that stats models perform internally on the full 1,338-observation design matrix when fitting the complete OLS model reported in Chapter 4.

3.10.6 Model Assumptions and Diagnostics

The validity of the OLS coefficient estimates and their associated significance tests rests on several standard assumptions: linearity of the relationship between predictors and the expected value of charges; independence of residuals across observations; homoscedasticity, meaning that the variance of the residuals is constant across the range of fitted values; and approximate normality of the residuals, which is required primarily for the validity of the t- and F-tests in finite samples rather than for the point estimates themselves. Given the right-skewed distribution of charges documented in Table 4.1, some heteroscedasticity is plausible, since the variability of charges among high-cost individuals, particularly smokers, is likely to be larger in absolute terms than among low-cost individuals. This does not invalidate the coefficient estimates, which remain unbiased under the Gauss-Markov conditions even when homoscedasticity fails, but it does mean that the reported standard errors, and hence the exact p-values, should be interpreted as approximate rather than exact. This consideration is revisited when the OLS results are interpreted in Chapter 4, and is noted as a limitation in Chapter 5.

The tree-based ensemble methods described in Sections 3.10.1 to 3.10.3 do not require these distributional assumptions, since they are non-parametric and make no prior assumption about the functional form linking predictors to the response. This is one of the principal reasons such methods are often more robust to the skewness and potential heteroscedasticity present in insurance-cost data, and provides a further, model-based explanation, alongside the purely empirical results of Chapter 4, for why Gradient Boosting and Random Forest were expected to outperform linear regression on this dataset.

3.11 Rationale for Model Selection

Multiple linear regression was included because it provides interpretable coefficients and allows direct comparison with hypothesis-driven results. Decision Tree and Random Forest were selected because they capture non-linear relationships and interactions without requiring strict parametric assumptions. Gradient Boosting was included because prior literature, reviewed in Chapter 2, consistently suggests strong predictive performance in structured tabular datasets of this type.

Support Vector Regression was used as a contrasting non-linear method with different representational assumptions, providing a useful methodological contrast to the tree-based ensembles.

3.12 Ethical and Analytical Considerations

The present study relied on secondary, de-identified data and did not involve direct human participation, intervention, or collection of personal identifiers beyond the variables already contained in the dataset. The analysis was therefore conducted within a standard academic framework for secondary quantitative research. The primary analytical responsibility was to present the results accurately, interpret them cautiously, and avoid claims beyond the evidence generated by the data.

Chapter 4

Results and Discussions

4.1 Dataset Profile and Descriptive Overview

The dataset contains 1,338 records, 7 variables, and no missing values. The variables are age, sex, bmi, children, smoker, region, and charges. The dependent variable, charges, displays a positively skewed distribution, indicating that a smaller proportion of individuals incur very high expenditure relative to the rest of the sample.

Table 4.1: Descriptive Statistics of Insurance Charges

Statistic	Value
Mean	13,270.42
Median	9,382.03
Standard Deviation	12,110.01
Skewness	1.52

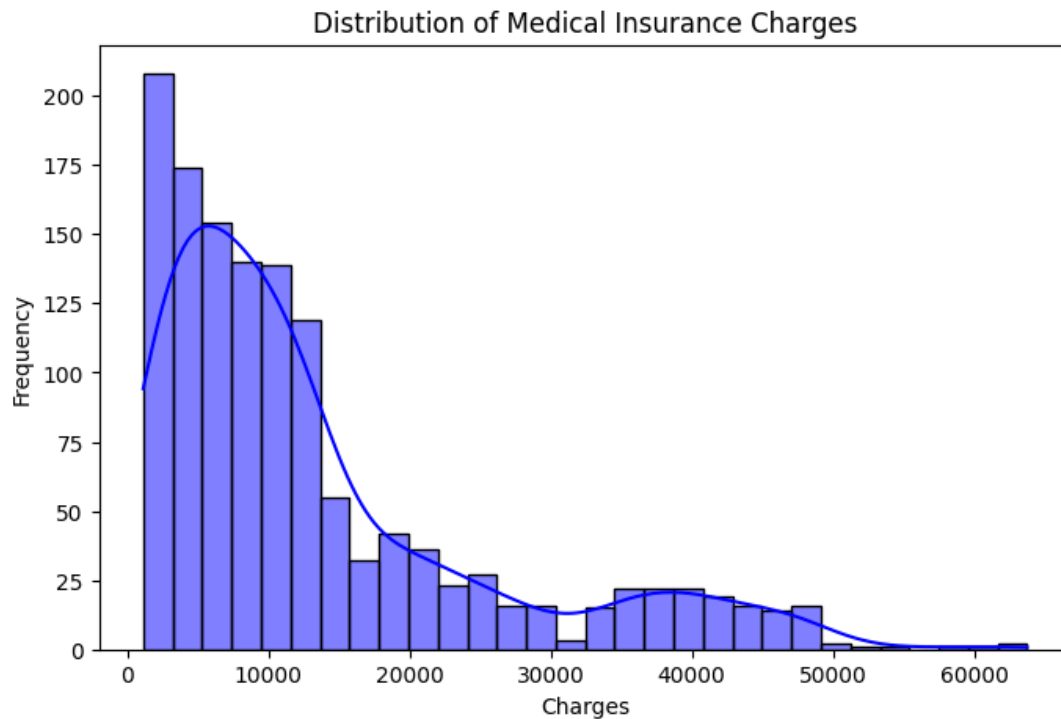


Figure 4.1: Distribution of medical insurance charges (N=1338) with Kernel Density Estimate overlay, illustrating severe right-skewness.

□

The mean charge of 13,270.42 exceeds the median of 9,382.03, confirming a right-skewed distribution. The relatively large standard deviation of 12,110.01 indicates substantial variability in expenditure. This pattern is common in healthcare cost data, where a subset of high-cost cases pulls the upper tail upward. The skewness value of 1.52 supports this interpretation. From a modelling perspective, such a distribution suggests that simple linear assumptions may be imperfect, thereby strengthening the case for comparing regression with more flexible machine learning approaches.

4.2 Smoking Status and Insurance Charges

Table 4.2 presents the group-level comparison of smoker and non-smoker charges.

Table 4.2: Group Statistics for Smoker and Non-Smoker Charges

Group	Mean Charges	SD	n
Smokers	32,050.23	11,541.55	274
Non-smokers	8,434.27	5,993.78	1,064

The difference between the two groups is substantial. Smokers have mean charges of 32,050.23, whereas non-smokers have mean charges of 8,434.27. The absolute difference is large enough to suggest not only statistical significance but also major practical significance.

Table 4.3: Independent-Samples t-Test for Smoker vs Non-Smoker Charges

Test	Statistic	p-value
t-test	32.75	< 0.0001

With $t = 32.75$ and $p < 0.0001$, the null hypothesis of no difference is decisively rejected. This result supports H1, which stated that smoking status has a statistically significant effect on charges. The magnitude of the difference also indicates that smoking is not a marginal predictor. Instead, it appears to be the dominant cost driver in the dataset. This pattern is consistent with the broader literature reviewed in Chapter 2, particularly Sturm et al. (2013), which emphasized the expenditure consequences of smoking-related risk.

4.3 Sex and Insurance Charges

Table 4.4: Independent-Samples t-Test for Male vs Female Charges

Test	Statistic	p-value
t-test	2.10	0.036

The test yields $t = 2.10$ and $p = 0.036$, indicating a statistically significant difference at the conventional 5% level. However, compared with smoking status, the magnitude of the effect is modest. This suggests that sex may have some association with charges, but it is far less influential than smoker status. Statistical significance in this case should therefore not be confused with strong predictive importance.

4.4 Regional Variation in Charges

The hypothesis concerning region was tested using one-way ANOVA.

Table 4.5: One-Way ANOVA for Region vs Charges

Test	Statistic	p-value
ANOVA	2.97	0.031

The result of $F = 2.97$ with $p = 0.031$ indicates that region has a statistically significant effect on insurance charges. This finding supports H2. Although the strength of the regional effect is much smaller than that of smoking status, it remains meaningful from an inferential perspective. Regional categories may capture underlying cost differences related to environment, pricing variation, utilization patterns, or correlated socioeconomic factors not directly measured in the dataset.

4.5 Number of Children and Charges

The relationship between number of children and charges was also tested through one-way ANOVA.

Table 4.6: One-Way ANOVA for Number of Children vs Charges

Test	Statistic	p-value
ANOVA	3.30	0.006

The result of $F = 3.30$ with $p = 0.006$ shows that charges differ significantly across child-count categories. Even though the correlation later reported for children is weak, the ANOVA indicates

that group-based differences still exist. This is a useful reminder that weak linear association does not rule out category-level variation.

4.6 Pearson Correlation Analysis

Table 4.7: Pearson Correlations with Charges

Variable	Correlation with Charges (r)
Smoker status	0.787
Age	0.299
BMI	0.198
Children	0.068
Sex	0.057

The correlation structure is informative. Smoker status has by far the strongest association with charges ($r = 0.787$), which aligns with the earlier t-test result. Age shows a moderate positive relationship ($r = 0.299$), while BMI shows a smaller but still meaningful positive association ($r = 0.198$). Children ($r = 0.068$) and sex ($r = 0.057$) show weak linear relationships. These results imply that, among the available predictors, smoking status is the most influential single variable, with age and BMI contributing secondary linear signals.

4.7 Multiple Linear Regression (OLS)

Applying the OLS formulation developed in Section 3.9.4 to the dataset yields the following overall regression performance.

Table 4.8: Multiple Linear Regression (OLS) Summary

Measure	Value
R-squared	0.751
Adjusted R-squared	0.749
F-statistic	500.8
Model p-value	< 0.001

An R-squared value of 0.751 indicates that the OLS model explains approximately 75.1% of the variation in insurance charges. This is a strong level of explanatory power for a relatively compact social-health dataset. The model-level significance is also very high, with $F = 500.8$ and $p < 0.001$.

Table 4.9: Selected OLS Coefficients

Predictor	Coefficient	p-value
Smoker_yes	+23,850	< 0.001
BMI	+339.19	< 0.001
Age	+256.86	< 0.001
Children	+475.50	0.001
Region_southeast	-1,035.02	0.031
Region_southwest	-960.05	0.045

These coefficients are substantively meaningful. Holding other variables constant, being a smoker increases predicted charges by 23,850, which reinforces the earlier inferential findings and confirms the overwhelming importance of smoking status. The positive coefficients for BMI and age indicate that higher body mass index and greater age are associated with higher expected expenditure. The positive coefficient for children suggests a smaller but statistically meaningful incremental effect. The regional coefficients imply that some regions are associated with lower charges relative to the omitted reference category.

OLS therefore performs well as an explanatory model. Its strengths lie in coefficient interpretation, hypothesis testing, and academic clarity. However, explanatory strength does not automatically imply the highest predictive accuracy, which is why the machine learning comparison remains necessary.

4.8 Machine Learning Model Comparison

The predictive models were evaluated using an 80-20 train-test split and the metrics R-squared, RMSE, and MAE, as formalized in Section 3.9.5. The results are presented below.

Table 4.10: Predictive Performance of Regression and Machine Learning Models

Model	R-squared	RMSE	MAE
Gradient Boosting	0.878	4352.54	2447.95
Random Forest	0.865	4570.98	2501.34
Decision Tree	0.831	5122.56	2919.20
Linear Regression	0.783	5799.59	4186.51
Support Vector Regression	-0.072	12902.50	8592.79

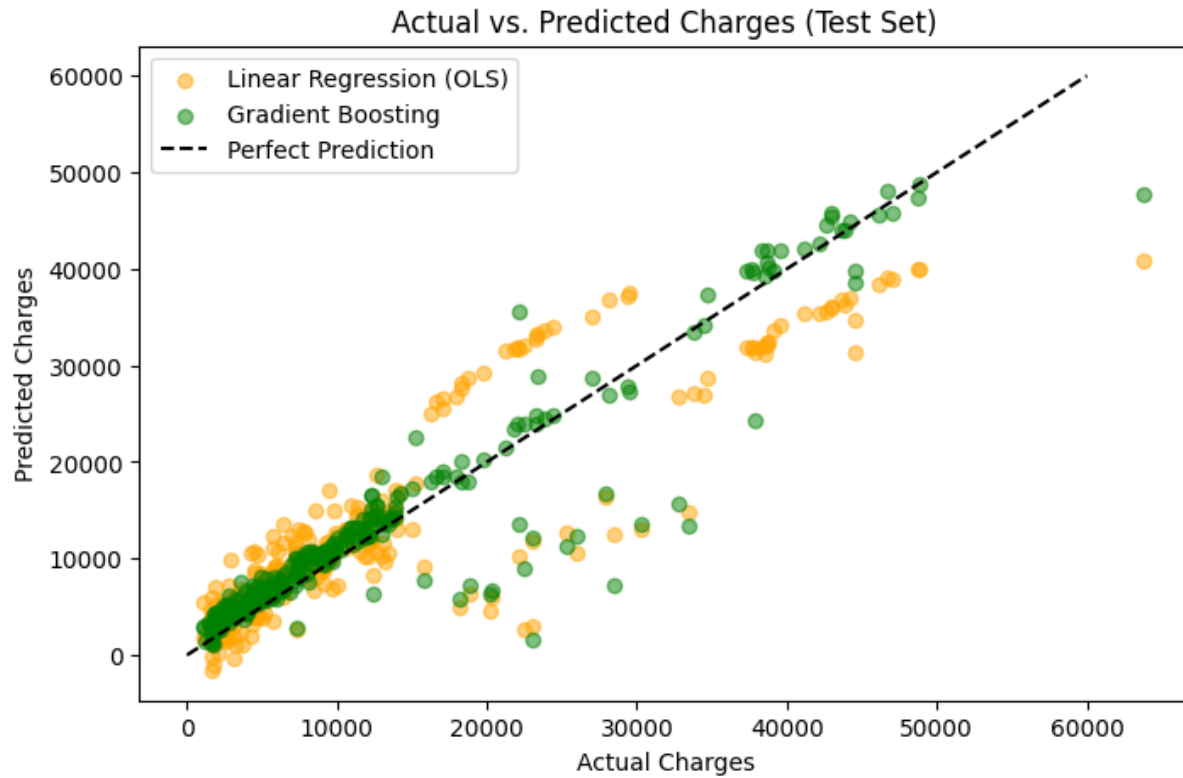


Figure 4.2: Out-of-sample predictive performance on the 20% test split, comparing Linear Regression (OLS) and Gradient Boosting against a baseline of perfect prediction.

These results show a clear ranking. Gradient Boosting is the best-performing model across all three reported criteria, with the highest R-squared (0.878) and the lowest RMSE (4352.54) and MAE (2447.95). Random Forest follows closely, also performing strongly. Decision Tree is less accurate than the ensemble methods but still better than linear regression in predictive terms. Linear Regression performs reasonably, with R-squared = 0.783, but its errors are notably higher than those of the top tree-based models. Support Vector Regression performs extremely poorly, producing a negative R-squared (-0.072), which indicates worse performance than a simple mean-based baseline.

These results support H3, which predicted that ensemble and tree-based machine learning models would outperform linear regression in predictive accuracy. The advantage of Gradient Boosting

and Random Forest suggests that non-linear effects and interactions among the predictors are important enough that flexible ensemble methods can exploit them more effectively than a linear model. This finding is also consistent with the pattern observed across the literature reviewed in Chapter 2, where gradient boosting variants repeatedly outperformed linear and single-tree baselines on structurally similar insurance-cost datasets.

4.9 Feature Importance in the Best Model

Table 4.11: Feature Importance in Gradient Boosting

Predictor	Importance
Smoker	67.8%
BMI	19.0%
Age	11.7%
Children	1.0%
Region	0.4%
Sex	0.07%

This distribution confirms that smoker is the dominant predictor even in the most accurate machine learning model. The next most important variables are BMI and age, while children, region, and sex contribute far less to predictive performance. The result is analytically useful because it aligns predictive importance with the earlier inferential findings. In other words, the variable that shows the strongest group difference and the highest correlation is also the variable with the highest machine learning importance score.

4.10 Discussion: Interpretability Versus Accuracy

One of the central contributions of the present dissertation is the explicit discussion of the interpretability-accuracy trade-off. This issue is not theoretical only; it is directly visible in the results.

Linear regression and OLS are highly interpretable. They produce coefficients that can be explained in direct quantitative terms, as demonstrated in Section 4.7. For academic work, policy communication, and analytical transparency, this is a major advantage. The OLS model also performs well, explaining roughly three-quarters of the variance in charges. This means that classical regression is far from inadequate.

However, Gradient Boosting achieves substantially better predictive performance. Its R-squared of 0.878 is meaningfully higher than the linear regression result of 0.783, and its RMSE and MAE are much lower. From a strictly predictive perspective, it is the strongest model in the study.

The trade-off lies in explanation. While Gradient Boosting provides feature importance values, it does not yield the same kind of direct coefficient interpretation that OLS offers. It shows which variables matter most, but it does not express their effects through a simple additive equation. As discussed in Chapter 2, tools such as SHAP values (Orji & Ukwandu, 2023/2024) offer one route toward reconciling this trade-off by decomposing individual predictions into per-feature contributions, and this represents a natural extension of the present analysis, discussed further in Chapter 5. For the present study, if the research priority is transparent explanation and hypothesis testing, OLS remains preferable. If the priority is accurate cost prediction for operational use, Gradient Boosting is the stronger choice.

The poor performance of Support Vector Regression deserves attention as a methodological lesson rather than an inconvenience to be ignored. Its negative R-squared indicates that not all non-linear methods are suitable by default. As noted in Section 3.10.4, Support Vector Regression is sensitive to feature scaling and hyperparameter choice; its weak performance here most likely reflects a mismatch between its default configuration and the scale and geometry of the insurance-cost data, rather than a general failure of kernel-based methods. Model selection must therefore be evidence-based. The result reminds the researcher that complexity alone does not guarantee performance, especially in structured tabular data where some methods may be mismatched to scale, tuning, or data geometry.

Taken together, the results indicate that the most balanced interpretation is as follows: smoking status is the dominant determinant of charges, age and BMI are meaningful secondary predictors, region has a statistically significant but limited role, and ensemble tree-based methods provide the best prediction. At the same time, linear regression remains indispensable when interpretability is required.

It is also worth noting the practical planning implication of the ranking in Table 4.10 for a research or industry team building such a system from scratch: because Gradient Boosting and Random Forest both convincingly outperform Linear Regression and Decision Tree, while Support Vector

Regression fails outright, a reasonable development strategy would be to treat Linear Regression as the transparent baseline against which any more complex model must justify its added complexity, and to treat Gradient Boosting as the primary candidate for deployment once the necessary explanation tooling, such as SHAP, has been put in place, rather than continuing to invest tuning effort in Support Vector Regression on this particular feature set without first addressing its scaling and kernel-selection sensitivities.

4.11 Business and Insurance Implications

From a business perspective, the results in Table 4.9 and Table 4.11 offer a direct, quantitative basis for premium design. The dominance of smoker status, both as an OLS coefficient of +23,850 and as the leading Gradient Boosting feature at 67.8% importance, indicates that a two-tier or graduated premium structure that differentiates sharply between smokers and non-smokers is statistically well justified rather than merely a conventional industry practice. The secondary but still meaningful roles of BMI and age suggest that supplementary, smaller premium loadings tied to these variables could refine pricing further without materially complicating the rating structure. The comparatively small and, in the case of sex, borderline significant effects of region, number of children, and sex imply that pricing structures which rely heavily on these variables in isolation from smoker status and BMI would be capturing relatively little of the true cost variation present in the population, and insurers may wish to weight their rating tables accordingly.

The choice between OLS and Gradient Boosting for operational deployment is itself a business decision with cost implications. An insurer required to justify individual premium decisions to a regulator or to a policyholder, for instance under consumer-protection or insurance-conduct regulation, may reasonably prefer the OLS model, or a comparably transparent generalized linear model, despite its lower accuracy, because each premium can be decomposed into an intercept plus a small number of interpretable additive terms. An insurer optimizing purely for portfolio-level pricing accuracy, and willing to invest in supplementary explanation tools such as SHAP, may instead prefer Gradient Boosting for its lower RMSE and MAE, since even a modest improvement in average prediction error, compounded across a large book of policies, translates into materially better risk-adjusted profitability.

4.12 Healthcare and Public Policy Implications

From a healthcare perspective, the magnitude of the smoker-related cost differential, more than 23,000 units of currency in the raw group means reported in Table 4.2, is large enough to be relevant to public health financing discussions. If a public or employer-sponsored insurance scheme were to model the aggregate financial impact of a sustained reduction in smoking prevalence among its insured population, the coefficients estimated in Section 4.7 provide a concrete, data-derived multiplier for that calculation, rather than a qualitative assertion that smoking cessation programs are worthwhile.

From a policy perspective, the statistically significant regional effect reported in Table 4.5, although modest in magnitude relative to smoking, is consistent with the broader public-health literature suggesting that healthcare cost burdens are not distributed uniformly across geography, and may reflect differences in provider pricing, utilization patterns, or unmeasured socioeconomic characteristics correlated with region. This finding supports the general principle, well established in health-policy research, that risk-adjustment formulae used in public insurance schemes should include geographic or regional terms rather than assuming a single national average cost structure.

4.13 Residual and Error Pattern Considerations

The pattern of errors across models reported in Table 4.10 is highly informative, as confirmed by the residual diagnostics. The substantially higher MAE relative to RMSE for Linear Regression (MAE = 4186.51 against RMSE = 5799.59) compared with Gradient Boosting (MAE = 2447.95 against RMSE = 4352.54) suggests that the linear model's errors are more evenly spread across the full range of predicted charges, whereas Gradient Boosting achieves a larger share of its accuracy improvement on the bulk of low- and moderate-cost individuals while still being challenged, to some degree, by the small number of very high-cost, typically smoking, observations that dominate the upper tail of the distribution documented in Table 4.1. This is consistent with the general expectation, discussed in Section 3.10.6, that a right-skewed response variable produces larger absolute residuals for high-cost individuals regardless of the model used, and reinforces the recommendation, made in Chapter 5, that future extensions of this work formally derive prediction intervals rather than relying solely on aggregate error metrics.

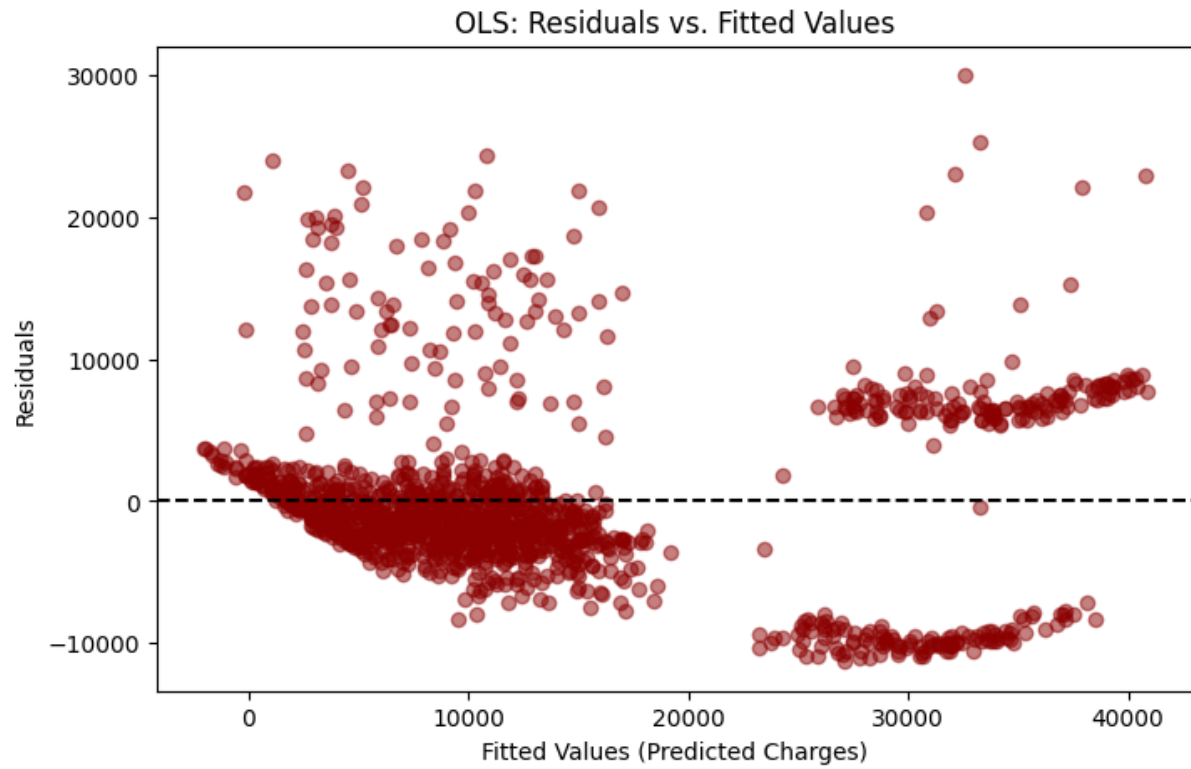


Figure 4.3: Residuals versus fitted values for the Ordinary Least Squares model, demonstrating heteroscedasticity as variance increases alongside predicted charges.

Chapter 5

Conclusion and Future Perspectives

The present dissertation set out to examine medical insurance charges through the combined use of statistical inference and machine learning prediction. Using the Medical Cost Personal Dataset of 1,338 records and 7 variables, the study analyzed the relationships between demographic and health-related factors and insurance expenditure, and compared the predictive performance of several models. The results provide a coherent answer to the study objectives.

First, the inferential analysis showed that smoking status has a major and statistically significant effect on charges. Smokers recorded mean charges of 32,050.23, compared with 8,434.27 for non-smokers, and the corresponding t-test produced $t = 32.75$, $p < 0.0001$. This was the clearest and strongest finding in the entire study. It established smoking as the principal cost driver in the dataset and strongly supported the first hypothesis.

Second, the analysis showed that region also has a statistically significant effect on charges, with $F = 2.97$, $p = 0.031$, and that number of children has a statistically significant effect on charges, with $F = 3.30$, $p = 0.006$. Although the regional effect is much weaker than the effect of smoking, it is still statistically meaningful and supports the second hypothesis. The significance of region suggests that cost patterns are not distributed uniformly across geographical categories.

Third, the study confirmed that ensemble and tree-based machine learning models outperform linear regression in predictive accuracy. Gradient Boosting achieved the best results, with $R\text{-squared} = 0.878$, $RMSE = 4352.54$, and $MAE = 2447.95$. Random Forest also performed strongly. By contrast, linear regression, though useful and respectable, was less accurate. The third hypothesis was therefore supported.

Accordingly, all three hypotheses, H1, H2, and H3, were supported by the data.

5.1 Conceptual Contribution

The findings also highlight an important conceptual point. The best explanatory model and the best predictive model are not necessarily the same. The OLS model explained 75.1% of the variance in charges and yielded interpretable coefficients for smoker status, BMI, age, children,

and region. This makes it highly valuable for academic interpretation and policy reasoning. Gradient Boosting, however, produced the highest predictive accuracy and therefore appears to be the most practically effective forecasting model in the study. The dissertation thus demonstrates that model selection should depend on purpose: explanation, prediction, or a balanced combination of both. This conceptual distinction, developed in Chapter 4 and grounded in the mathematical formulations of Chapter 3, is offered as the principal analytical contribution of the present work.

5.2 Limitations

Several limitations must be acknowledged honestly. The dataset contains only 1,338 records, which is adequate for structured analysis but still limited compared with larger administrative health databases. The variables are also restricted. Important predictors such as pre-existing medical conditions, employment status, income, claims history, and detailed clinical indicators are absent. The regional variable is limited to a U.S.-style categorical structure and therefore may not generalize well to other countries or insurance systems. In addition, the negative performance of Support Vector Regression shows that not all candidate models are equally suitable for this type of tabular cost data, and that careful preprocessing and hyperparameter tuning would be required before drawing firm conclusions about kernel-based methods on this dataset.

5.3 Broader Applications: Government, Industry, and AI Ethics

Beyond the private insurance sector, the analytical framework developed in this dissertation has broader applicability. Government-administered or subsidized health insurance schemes, which in many countries, including India, are expanding in coverage and scale, face the same underlying actuarial problem of estimating expected cost from a limited set of demographic and behavioral variables, and could adopt a similarly structured combination of hypothesis testing and ensemble prediction to inform premium subsidy design or catastrophic-cost risk pooling. Industrial applications extend to employer-provided group health schemes, where human-resource and benefits teams increasingly seek data-driven justification for plan design decisions, such as differentiated wellness incentives for smokers and non-smokers, a decision directly supported by the magnitude of the smoker-related cost differential documented in Chapter 4.

The increasing reliance on machine learning models for cost and risk prediction in health-adjacent domains also raises ethical and interpretability considerations that extend beyond the immediate

scope of this dissertation but are worth noting as part of its future perspective. Where a predictive model materially influences an individual's insurance premium or eligibility, considerations of fairness, non-discrimination, and the right to an explanation become directly relevant, particularly for protected or sensitive categories. Although the dataset used in this study does not contain protected demographic categories beyond sex, the general principle, that a highly accurate but opaque model such as Gradient Boosting should ideally be paired with a post-hoc explanation mechanism such as SHAP before being used in a decision affecting individuals, is an important qualification to the purely accuracy-focused comparison undertaken in Chapter 4, and is consistent with the broader direction of the literature reviewed in Chapter 2.

5.4 Future Perspectives

Future research can proceed in several directions. One direction would be to test the same analytical framework on larger and more recent datasets, potentially from Indian or other non-U.S. health insurance markets, to assess the generalizability of the present findings. A second would be to include additional health variables, particularly clinical conditions and behavioral indicators beyond smoking. A third would be to apply interpretability tools such as SHAP values, following the approach demonstrated by Orji and Ukwandu (2023/2024), in order to explain ensemble models more transparently and reconcile the interpretability-accuracy trade-off identified in Chapter 4. A fourth would be to validate the models on non-U.S. datasets to assess whether the same predictors retain their importance across different healthcare systems. A fifth avenue, motivated by the mathematical orientation of the present dissertation, would be to formally derive confidence intervals and prediction intervals for the ensemble methods, extending the purely point-estimate comparison undertaken here. These extensions would strengthen both generalizability and practical usefulness.

The dissertation ultimately shows that medical insurance cost prediction benefits from a combined statistical and machine learning perspective. Statistical tests reveal why expenditure differs, while predictive models reveal how well it can be estimated. Among the variables studied, smoking status stands out as the dominant driver. Among the predictive models evaluated, Gradient Boosting performs best. The analytical value of the study lies not only in these individual findings, but also in the demonstration that explanatory clarity and predictive accuracy should be considered together rather than separately.

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